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Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

USE OF THE OBSTETRIC HIGH OBSERVATION UNIT AND REFERRAL OF OBSTETRIC PATIENTS FOR INTENSIVE CARE

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1. Introduction and Purpose

Occasionally parturients need high dependency or intensive care antenatally or postnatally, for co-morbidity which may or may not be related to their pregnancy.

2. Scope

This guideline is intended as a reference for trainee anaesthetists, obstetricians and midwives to help decide which parturients need critical care.

3. Points to consider

3.1 Provision of Level 1 & 2 Obstetric Critical Care

- Obstetric patients requiring Level 1 critical care should be managed in the Obstetric High Observation Unit (OHOU) ie. room 8 on Delivery suite by a midwife with Enhanced Maternal Care skills[1] (EMC Midwife)

- Obstetric patients requiring pre-delivery Level 2 critical care should also be managed in the Obstetric High Observation Unit (OHOU). Care should be delivered by an EMC Midwife; with additional critical care support being closely provided by the resident duty obstetric anaesthetist.

- Obstetric patients requiring antenatal (excluding pre-delivery) & postnatal Level 2 critical care should be managed in the main critical care unit. Care should be provided jointly by a critical care nurse and an EMC Midwife, along with shared medical care between the obstetric & critical care teams. If postnatal, the patient should be provided a cubicle to facilitate baby access. If no critical care bed is available then manage the patient in OHOU as per 'pre-delivery Level 2 critical care'.

3.2 Obstetric High Observation Unit

- Facilities are available in the OHOU to monitor ECG, pulse oximetry, invasive arterial blood pressure and central venous pressure via a portable Mindray monitor (usually kept in the labour ward store room).

- The patients vital signs should all be recorded on a critical care observation chart.

- The designated OHOU midwife must possess Enhanced Maternal Care (EMC) skills to be competent in the care of the critically ill obstetric patient; and in the care of both arterial and CVP lines. Therefore, each labour ward midwifery shift must have a designated EMC competent midwife.

- One to one EMC Midwifery care is mandatory for OHOU patients.

- Patients who need critical care monitoring but do not need invasive mechanical ventilation or airway protection by intubation should be transferred to the obstetric high observation unit.

- Examples of obstetric indications for Level 1 & 2 critical care include :

  Level 1

  - Patients requiring additional monitoring or intervention, or step down from higher level of care e.g.

    i. Risk of haemorrhage

    ii. Oxytocin infusion

    iii. Mild pre-eclampsia on oral anti-hypertensives / fluid restriction etc

    iv. Patients with medical conditions such as congenital heart disease or a diabetic on an insulin infusion

  Level 2

  - Patient requiring single organ support e.g.

    i. Basic Respiratory Support

    - 50% or more oxygen via face-mask to maintain oxygen saturation
    - Continuous Positive Airway Pressure (CPAP), Bi-Level Positive Airway Pressure (BIPAP)

    ii. Basic Cardiovascular Support

    - Intravenous anti-hypertensives, to control blood pressure in pre-eclampsia
    - Arterial line used for pressure monitoring or sampling
    - CVP line used for fluid management and CVP monitoring to guide therapy

    iii. Neurological Support

    - Magnesium infusion to control seizures

3.3 Intensive Care Unit referral (Level 3 Care)

- Patients should be referred to the intensive care unit when they require level 3 critical care.

- The case should always first be discussed with the intensive care consultant, who then decides whether the patient warrants admission to ITU.

- Care should be provided by a critical care nurse supported by a midwife, along with shared medical care between the obstetric & critical care teams.

- If postnatal, the patient should be provided a cubicle to facilitate baby access

- The criteria for level 3 care includes :

    i. Advanced Respiratory Support

        - Need for invasive mechanical ventilation

    ii. Support of two or more organ systems

        - Renal support and Basic Respiratory Support

        - Basic Respiratory Support / Basic Cardiovascular Support and an additional organ supported

    iii. Advanced Cardiovascular Support

        - Simultaneous use of at least two intravenous, anti-arrhythmic / antihypertensive / vasoactive drugs, one of which must be a vasoactive drug

        - Need to measure and treat cardiac output

    iv. Management of acute fulminant hepatic failure, e.g. from HELLP syndrome or acute fatty liver (may require advice from the regional liver unit)

3.4 Transfer to the Intensive Care Unit

- The patient should be transferred from delivery suite to the intensive care unit using full monitoring ie. ECG, pulse oximetry, NIBP / IABP and a capnograph (for mechanically ventilated patients).

- The patient should be accompanied by the anaesthetist, ODP and a midwife.

- Full inter hospital transfer equipment eg. laryngoscope, ET tubes and anaesthetic drugs etc.(including cardiac arrest box) transport ventilator / Ambu bag, oxygen cylinder, syringe drivers and portable suction should accompany the patient.

4. References

1. Care of the critically ill woman in childbirth; enhanced maternal care (2018). OAA, RCoA, RCOG,RCM,FICM & ICS

2. The Royal College of Anaesthetists.Providing equity of critical and maternity care for the critically ill pregnant or recently pregnant woman (2011) RCoA, London.